Full-dataset LightGBM Report

Full-dataset LightGBM ablation

Dataset: all 2,880 rows

Validation: 70/30 stratified holdout + repeated grouped CV on training split

Feature sets compared: embeddings only vs embeddings plus region and Dynamic World context

Best held-out feature set by target:

- S1_VH: Embeddings only with R2=0.942, RMSE=1.182, MAE=0.858, r=0.971
- S1_VV: Embeddings only with R2=0.902, RMSE=1.255, MAE=0.825, r=0.951
- S1_VV_div_VH: Embeddings + region + Dynamic World with R2=0.698, RMSE=1.155, MAE=0.798, r=

Interpretation:

- S1_VV: Embeddings only leads (0.902 vs 0.901).
- S1_VH: Embeddings only leads (0.942 vs 0.941).
- VV/VH ratio: Embeddings + region + Dynamic World leads (0.698 vs 0.698).
- Stability metrics summarize repeated grouped-CV over three random seeds.

Held-out Performance by Feature Set

target	feature_set	cv_r2_mean	cv_rmse_mean	stability_r2_mean	stability_rmse_me	r2	rmse	mae	pearson_r
S1_VH	Embeddings only	0.931	1.238	0.934	1.246	0.942	1.182	0.858	0.971
S1_VH	Embeddings + region + Dyn	0.930	1.256	0.933	1.257	0.941	1.199	0.881	0.970
S1_VV	Embeddings only	0.903	1.155	0.907	1.149	0.902	1.255	0.825	0.951
S1_VV	Embeddings + region + Dyn	0.902	1.164	0.904	1.168	0.901	1.260	0.825	0.951
S1_VV_div_VH	Embeddings + region + Dyn	0.696	1.050	0.715	1.059	0.698	1.155	0.798	0.836
S1_VV_div_VH	Embeddings only	0.696	1.045	0.713	1.061	0.698	1.156	0.808	0.836

The holdout test set is shared across the two feature-set ablations.

Repeated Grouped-CV Stability

target	feature_set	r2_mean	r2_std	rmse_mean	rmse_std	mae_mean	mae_std
S1_VH	Embeddings only	0.934	0.012	1.246	0.072	0.929	0.039
S1_VH	Embeddings + region + Dyna	0.933	0.010	1.257	0.077	0.946	0.050
S1_VV	Embeddings only	0.907	0.016	1.149	0.062	0.831	0.037
S1_VV	Embeddings + region + Dyna	0.904	0.016	1.168	0.070	0.846	0.043
S1_VV_div_VH	Embeddings only	0.713	0.071	1.061	0.051	0.793	0.026
S1_VV_div_VH	Embeddings + region + Dyna	0.715	0.069	1.059	0.060	0.795	0.041

Each row summarizes 12 grouped validation folds: 4 folds across 3 random seeds.

Regional S1_VV Diagnostics

region	feature_set	r2	rmse	mae	pearson_r	n_test
amazon_forest	Embeddings only	0.936	1.125	0.784	0.967	216
california_coast	Embeddings only	0.923	1.027	0.740	0.961	216
iowa_ag	Embeddings only	0.851	1.302	0.751	0.930	216
sf_bay_urban	Embeddings only	0.871	1.512	1.026	0.936	216
amazon_forest	Embeddings + region + Dynam	0.935	1.129	0.778	0.967	216
california_coast	Embeddings + region + Dynam	0.922	1.029	0.750	0.961	216
iowa_ag	Embeddings + region + Dynam	0.839	1.357	0.789	0.923	216
sf_bay_urban	Embeddings + region + Dynam	0.878	1.474	0.981	0.940	216

Region-level diagnostics on the held-out test set.

Land-Use S1_VV Diagnostics

dw_label_name	feature_set	r2	rmse	mae	pearson_r	n_test
bare	Embeddings only	0.692	1.202	0.837	0.847	96
built	Embeddings only	0.741	2.070	1.227	0.876	96
crops	Embeddings only	0.769	1.006	0.772	0.878	96
flooded_vegetation	Embeddings only	0.930	0.744	0.528	0.965	96
grass	Embeddings only	0.863	0.960	0.697	0.931	96
shrub_and_scrub	Embeddings only	0.781	1.366	0.902	0.889	96
snow_and_ice	Embeddings only	0.897	1.020	0.737	0.948	96
trees	Embeddings only	0.861	0.894	0.715	0.930	96
water	Embeddings only	0.842	1.500	1.011	0.923	96
bare	Embeddings + region + Dynam	0.699	1.189	0.826	0.849	96
built	Embeddings + region + Dynam	0.732	2.108	1.233	0.872	96
crops	Embeddings + region + Dynam	0.782	0.978	0.724	0.885	96
flooded_vegetation	Embeddings + region + Dynam	0.917	0.810	0.559	0.958	96
grass	Embeddings + region + Dynam	0.858	0.978	0.691	0.929	96
shrub_and_scrub	Embeddings + region + Dynam	0.788	1.345	0.894	0.893	96
snow_and_ice	Embeddings + region + Dynam	0.898	1.016	0.761	0.948	96
trees	Embeddings + region + Dynam	0.847	0.939	0.745	0.923	96
water	Embeddings + region + Dynam	0.850	1.462	0.988	0.931	96

Land-use diagnostics on the held-out test set.

Feature Importance Summary

Top 5 features by permutation importance

Embeddings only

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S1_VV: A25 (0.120), A27 (0.089), A49 (0.063), A17 (0.039), A34 (0.030)
S1_VH: A25 (0.269), A27 (0.051), A49 (0.030), A19 (0.025), A34 (0.024)
S1_VV_div_VH: A25 (0.106), A24 (0.063), A39 (0.038), A63 (0.037), A30 (0.031)
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Embeddings + region + Dynamic World

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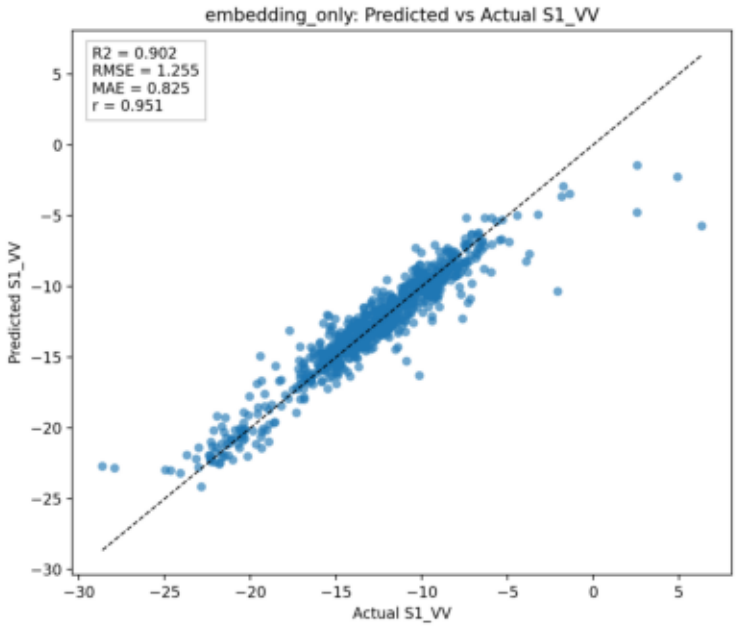
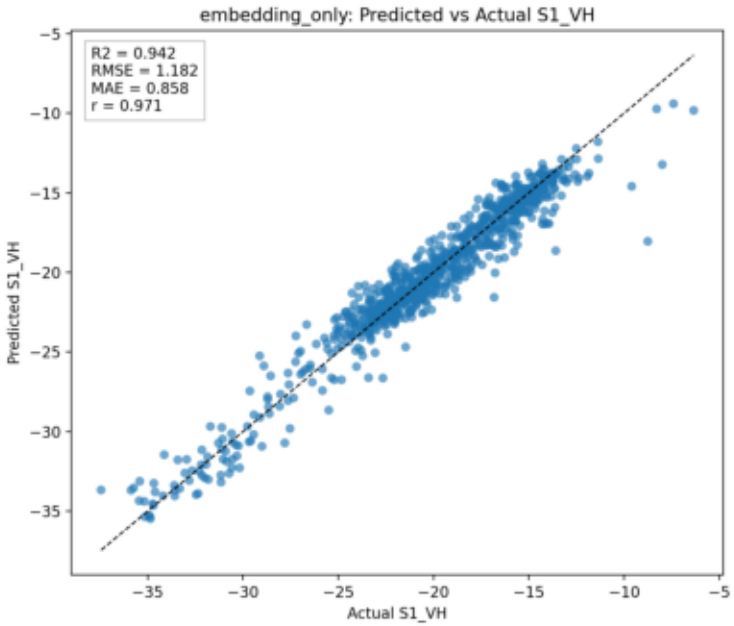
S1_VV: water (0.113), A27 (0.084), A49 (0.055), A25 (0.041), A17 (0.037)
S1_VH: water (0.134), A25 (0.112), A27 (0.050), bare (0.031), A34 (0.025)
S1_VV_div_VH: A25 (0.066), A39 (0.049), A63 (0.041), A24 (0.041), A30 (0.027)
```

Predicted vs Actual: Embeddings Only

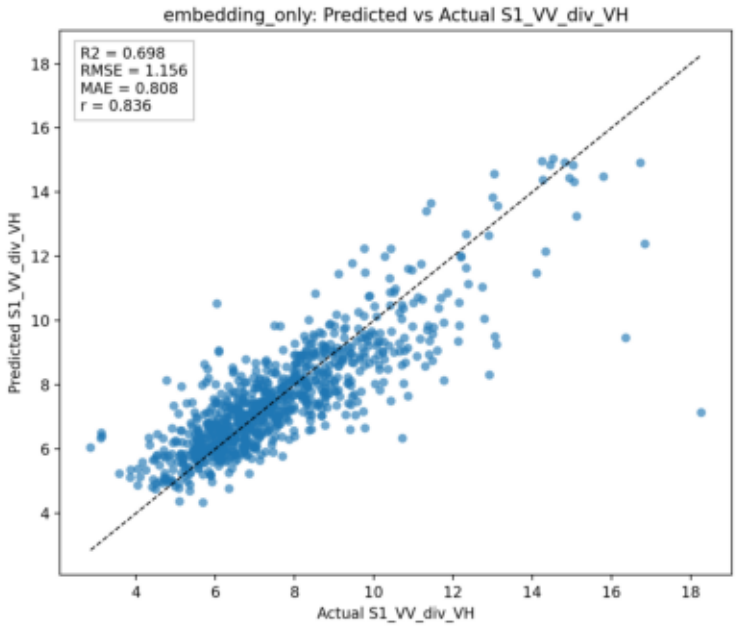
predicted vs actual embedding only S1 VV



predicted vs actual embedding only S1 VH

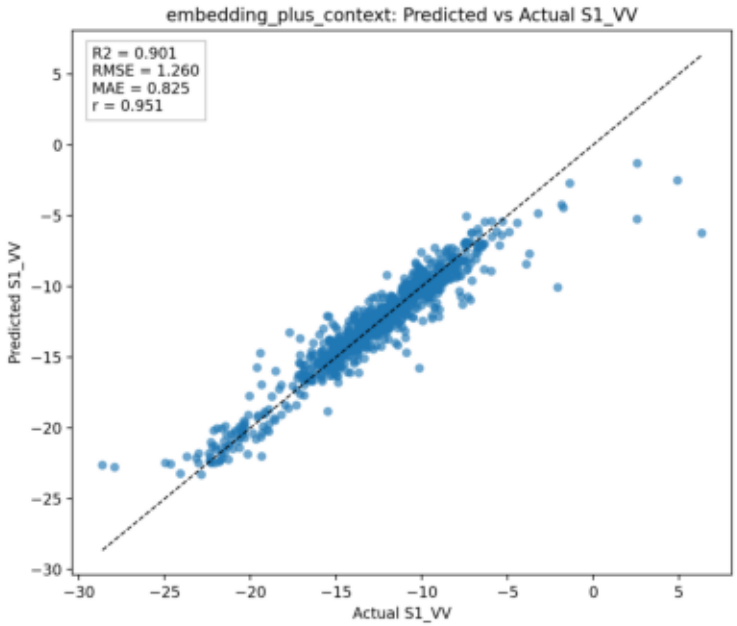


predicted vs actual embedding only S1 VV div VH

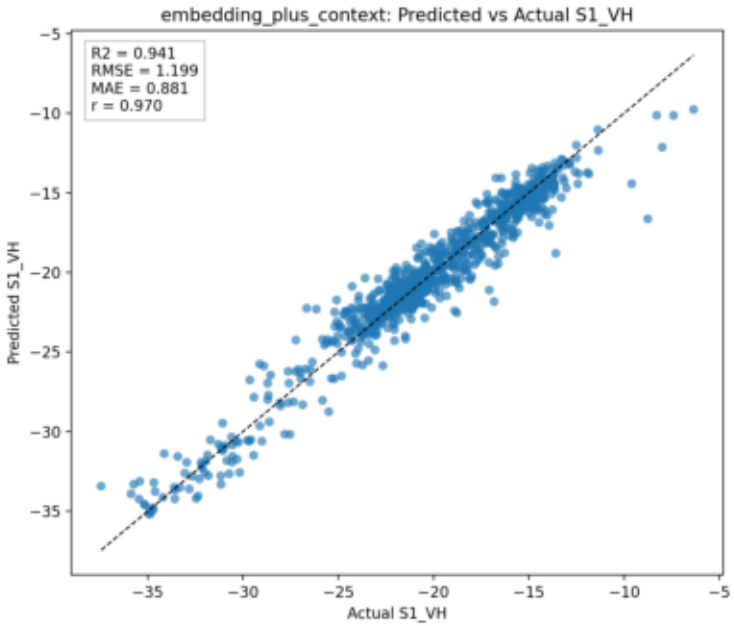


Predicted vs Actual: Embeddings + Context

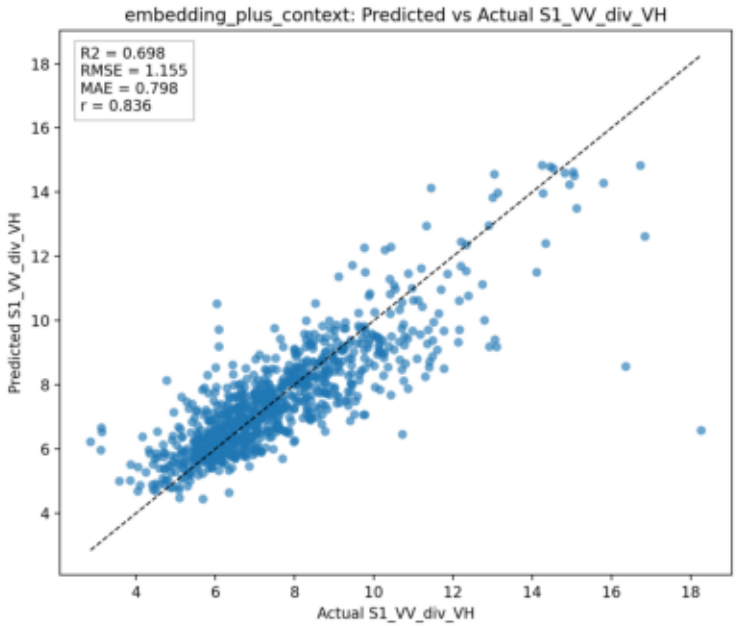
predicted vs actual embedding plus context S1 VV



predicted vs actual embedding plus context S1 VH

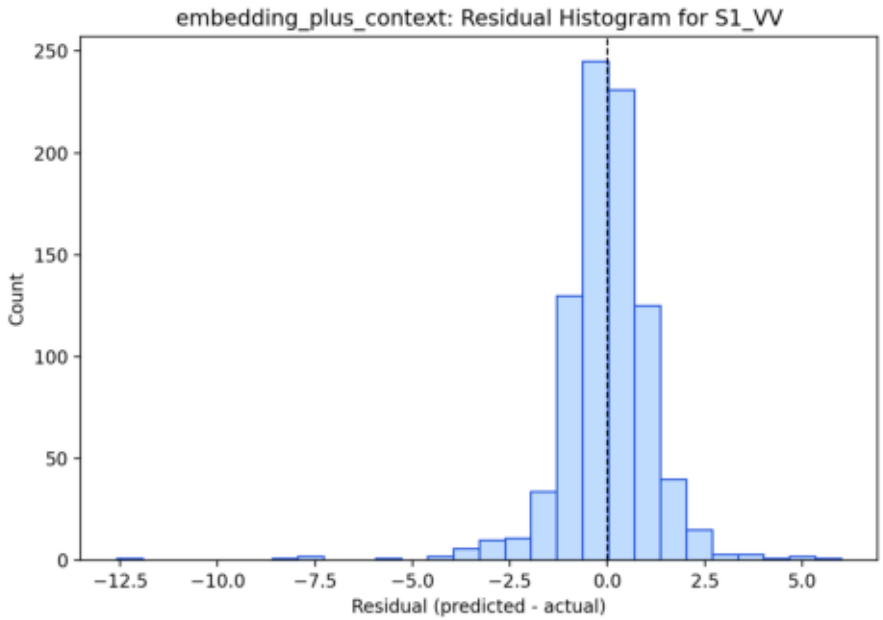


predicted vs actual embedding plus context S1 VV div VH

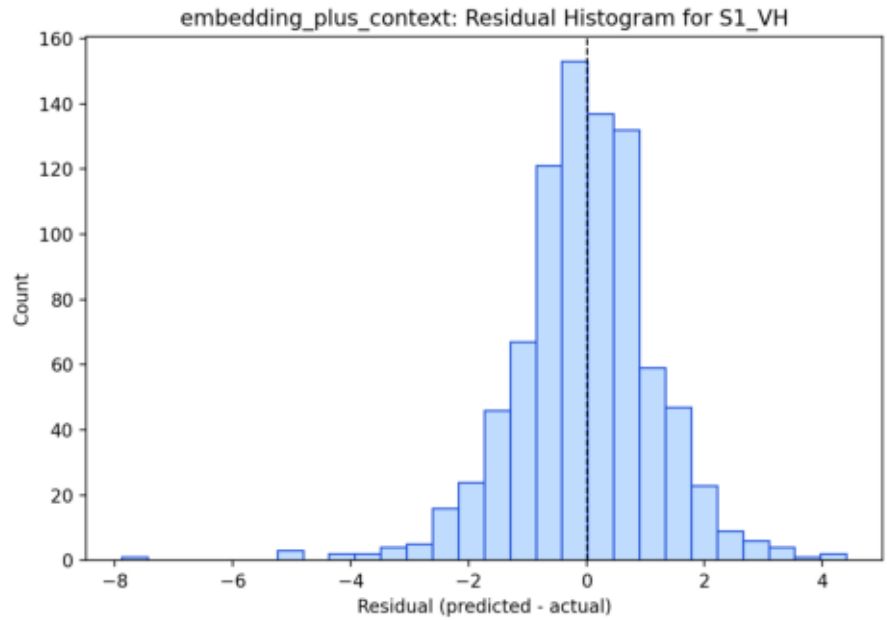
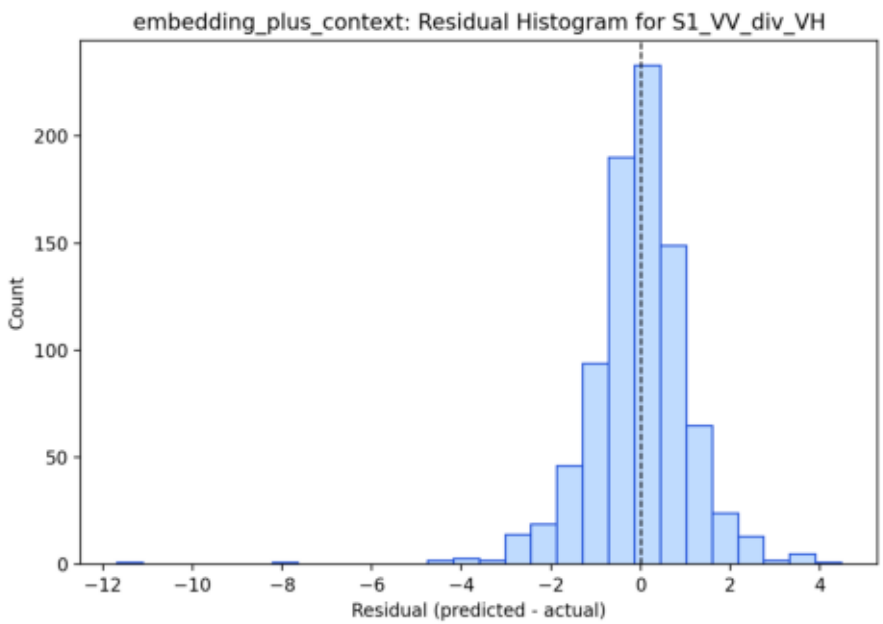


Residual Histograms: Embeddings + Context

residual histogram embedding plus context S1 VV



residual histogram embedding plus context S1 VH



residual histogram embedding plus context S1 VV div VH